13.0 CONDUCT OF OPERATIONS

13.1 ORGANIZATIONAL STRUCTURE OF APPLICANT

13.1.1 MANAGEMENT AND TECHNICAL SUPPORT ORGANIZATION

This section provides information concerning corporate organization, functions, and responsibilities; participation in the facility design; design review; design approval; construction management; testing; and operation of the plant. The corporate nuclear operations organization and the Southern Nuclear Operating Company (SNC) plant organization are responsible for directing activities at VEGP. The organizations described in chapter 13 support and report to nuclear operations for assigned activities.

13.1.1.1 Design and Operating Responsibilities

The following paragraphs summarize the degree to which design, construction, and preoperational activities were accomplished and describe the specific responsibilities and activities relative to technical support for operations.

13.1.1.1.1 Design and Construction Activities

13.1.1.1.1.1 Principal Site-Related Engineering Work. Principal site-related work (such as meteorology, geology, seismology, hydrology, and demography) is described in chapter 2. The VEGP preoperational monitoring program is described in the environmental report; this program established a preoperational baseline from which to evaluate future monitoring of environmental effects.

13.1.1.1.1.2 Design of Plant and Auxiliary Systems. Units 1 and 2 engineering and construction are complete.

13.1.1.1.1.3 Site Layout with Respect to Environmental Effects and Security Provisions. Site layout with respect to environmental effects is described in chapter 2. Site security with respect to plant geographical layout and equipment is described in the security plan.

13.1.1.1.1.4 Development of Safety Analysis Reports. Overall coordination of the preparation of the original Final Safety Analysis Report (FSAR) and its annual updates was assigned to Southern Company Services (SCS) Nuclear Support and Quality Assurance. Preparation of the individual sections was assigned to the cognizant technical groups within Bechtel, Westinghouse, SCS, and Georgia Power Company (GPC). As of March 22, 1997, SNC, as the exclusive operating licensee, reviews and approves the FSAR and annual updates and maintains overall responsibility for the FSAR. As a result of the consolidation of SCS and

SNC nuclear expertise, and in addition to being the licensee, SNC also serves as its own architect/engineer and performs the functions previously performed by SCS.

13.1.1.1.1.5 Review and Approval of Material and Component Specifications. Project specifications for safety-related equipment are reviewed in accordance with the quality assurance program as described in chapter 17.

13.1.1.1.1.6 Procurement of Materials and Equipment. Procurement of Unit 2 materials and major engineered equipment is complete.

13.1.1.1.2 Preoperational Activities

13.1.1.1.2.1 Development of Human Engineering Design Objectives and Design Phase Review of Proposed Control Room Layouts. The VEGP control room was designed using a reduced-size control board and GPC operator input into the control board control configurations. The design incorporated the human factor design criteria available at that time. An independent evaluation on human factor design has been performed on a mockup of the control room. A detailed discussion of control room design review and human engineering factors is described in chapter 18.

Design recommendations based on the mockup review were incorporated in the control board design or deferred for further evaluation. A detailed design review was conducted on the as-built control room and used the plant simulator.

13.1.1.1.2.2 Development and Implementation of Staff Recruiting and Training Programs. The operating staff is described in subsection 13.1.2. Recruiting of personnel to fill these positions started in 1977. The initial plant staffing is complete. Training programs have been developed for this facility and are described in section 13.2.

13.1.1.1.2.3 Development of Plans for Initial Testing. The nuclear plant general manager (NPGM) was responsible for all aspects of the initial test program of the VEGP. As part of his responsibilities, the NPGM directed the development of the startup manual.

The startup manual defined the startup organization, defined the responsibilities of involved organizations and personnel, delineated the qualifications necessary for startup personnel, and contained the administrative controls necessary for the implementation of the initial test program.

The administrative controls, qualification for testing personnel, and other required procedures for conducting that part of the initial test program after fuel load was included in the plant procedure manual or startup test procedures manual.

13.1.1.1.2.4 Development of Plant Maintenance Programs. The work force assigned to the VEGP provided qualified maintenance personnel prior to initial fuel loading.

Structures, systems, and components that prevent or mitigate the consequences of postulated accidents that could cause undue risk to the health and safety of the public are maintained in accordance with the quality assurance program.

The maintenance staff is sized to perform the routine and preventive maintenance workload. The staff is supplemented by outside contractors as deemed appropriate by plant management. Maintenance is performed under the direction of cognizant team leaders or assistant team leaders and in accordance with accepted work practices.

The scope and frequency of the preventive maintenance considers such factors as past experience with similar equipment, the manufacturer's recommendations, and importance of the equipment to plant operations or nuclear safety. Records are kept to establish the maintenance history of major safety-related equipment. Maintenance and repairs are performed by qualified personnel in accordance with written work orders, maintenance procedures, standing orders, vendor technical manuals, and/or applicable codes and regulations. Qualified maintenance personnel possess the skills to perform work without detailed written procedures. Except for emergencies, maintenance work is preplanned. Training meetings are held to foster safety awareness and quality of workmanship.

13.1.1.1.3 Technical Support for Operations

SNC is responsible to GPC for assuring the availability of, for providing, or for securing adequate technical support for the VEGP. SNC may use the services of SNC, Bechtel, Westinghouse, and others as appropriate. Portions of the technical support from these organizations may be fulfilled through the use of outside contractors.

13.1.1.1.4 SNC Technical Support

The SNC technical support organization includes all major design disciplines and contracts for outside specialty technical support when additional expertise is needed and when major surges in manpower needs occur. SNC and others provide for various assigned support activities, which normally include the following:

A. Architect-engineering services as required for the design-engineering of plant modifications, including maintenance-related design changes, plant improvement-related design changes, and design changes or major plant additions as a result of new regulatory requirements and commitments. These services include both conceptual and detailed design, issue and maintenance of design drawings and specifications, incorporation of as-built notices, procurement, related quality assurance functions, etc.

B. Design-related evaluation and analysis.

C. Site assistance as required on 10 CFR 50.59 evaluations and analyses which are not directly related to design; e.g., with respect to operational requirements, technical specification changes, etc.

D. Engineering and maintenance support to ensure reliable operation of components.

E. Licensing-related support for NRC interface and review of generic issues and plant changes.

F. Nuclear fuel procurement and contract management.

G. Nuclear fuel support, including core analysis.

H. Selected generic issues and projects.

I. Plant chemistry support.

13.1.1.2 Organizational Arrangement

13.1.1.2.1 Nuclear Operations Organization

The nuclear operations organization, under the supervision of the President/CEO, has direct responsibility for the operation and maintenance of Southern Company's nuclear plants. The nuclear operations organization consists of the plant operating staffs and corporate management, planning and performance and quality assurance. Engineering support is provided primarily by the corporate and site engineering organizations as described herein.

As shown on figures 13.1.1-1 and 13.1.1-2, the president and CEO, executive vice president/CNO, executive vice president operational readiness & site integration, and vice president-Vogtle provide line management direction for the operation of the plant.

The structure of the nuclear operations organization is described in the following paragraphs. Portions of the SNC Fleet Operations Support, Engineering, General Counsel and External Affairs, and Human Resources organizations are also described in the following paragraphs.

13.1.1.2.1.1 President and CEO. The president/CEO is responsible for all aspects of operation of Southern Company's nuclear plants including employment decisions. The president/CEO is also responsible for all technical and administrative support activities provided by SNC and nonaffiliated contractors. The president/CEO directs the executive vice president/chief nuclear officer (CNO); the vice president and general counsel; the vice president-regulatory affairs; the vice president comptroller, treasurer, and secretary; and the human resources director in fulfillment of their responsibilities. The president/CEO reports to the Board of Directors with respect to all matters.

13.1.1.2.1.2 Executive Vice President/CNO. The executive vice president/CNO reports to the president/CEO. He/she is responsible for the safe, reliable, and efficient operation of VEGP, the Joseph M. Farley Nuclear Plant (FNP), and the Edwin I. Hatch Nuclear Plant (HNP). The executive vice president/CNO directs the vice president-Vogtle, the vice president-Hatch, the vice president-Farley, the vice president-nuclear oversight, the vice president-engineering, and the vice president-fleet operations support.

13.1.1.2.1.2.1 Vice President-Nuclear Oversight. The vice president-nuclear oversight reports to the executive vice president/CNO. This individual is responsible for SNC nuclear oversight activities using staffs located at corporate headquarters and at each of the operating plants. Activities include performance of independent assessments and verification of QA program implementation in accordance with regulatory commitments. The vice president-nuclear oversight directs the nuclear oversight managers at the three SNC sites and the corporate office.

13.1.1.2.1.2.2 Nuclear Oversight Managers. The responsibilities of the nuclear oversight managers are described in the SNC Quality Assurance Topical Report (QATR) applicable to operation-phase activities at SNC-operated plants. The responsibilities of the nuclear oversight managers are those described for the senior manager responsible for QA in the QATR. In addition, the nuclear oversight managers are responsible for managing independent assessments. The nuclear oversight managers report to the vice president-nuclear oversight. The nuclear oversight managers are responsible for verifying implementation of the QA program for design, testing, operation, and maintenance in accordance with the requirements of the QATR. The nuclear oversight organization is composed of a staff at the corporate headquarters and the site of each SNC-operated plant.

The nuclear oversight organization does not provide technical support (as defined in subsection 13.1.1 of Regulatory Guide 1.70) for the operation of VEGP. The activities of the nuclear oversight organization are fully described in the QATR.

13.1.1.2.1.2.3 Vice President-Fleet Operations Support. The vice president-fleet operations support reports to the executive vice president/CNO and is responsible for identifying and resolving fleet issues, and utilizing trends, operating experience, and industry best practices to improve fleet performance. The vice president-fleet operations support directs the fleet operations manager, the fleet maintenance manager, the fleet work management director, the fleet training manager, the fleet security director, and the environmental affairs, chemistry and radiological services manager. In addition, the vice president-fleet operations support directs the manager-information technology.

13.1.1.2.1.2.3.1 Fleet Operations Manager. The fleet operations manager is responsible for overall governance, oversight, and support of nuclear plant operations and related functions. The fleet operations manager establishes policy level guidance, provides strategic direction to plant operations departments regarding operating practices and standards, evaluates programs for conformance to industry best practices, and drives performance improvements where needed.

13.1.1.2.1.2.3.2 Fleet Maintenance Manager. The fleet maintenance manager is responsible for overall governance, oversight, and support of nuclear plant maintenance and related functions. The fleet maintenance manager establishes policy level guidance, provides direction to plant maintenance departments regarding maintenance practices, and drives performance improvements where needed.

13.1.1.2.1.2.3.3 Fleet Work Management Director. The fleet work management director is responsible for overall governance, oversight, and support of refueling outage planning and execution and the online work control process. The fleet work management director establishes policy level guidance, provides direction to plant staffs regarding online work controls and outage practices and standards, evaluates programs for conformance to industry best practices, and drives performance improvements where needed.

13.1.1.2.1.2.3.4 Fleet Training Manager. The fleet training manager is responsible for overall governance, oversight, and support of training and qualification related functions. The fleet

training manager establishes policy level guidance, provides direction to nuclear plant training departments regarding practices and standards, evaluates programs for conformance to industry best practices, and drives improvements where needed.

13.1.1.2.1.2.3.5 Environmental Affairs, Chemistry, and Radiological Services Manager. The environmental affairs, chemistry and radiological services manager reports to the vice president-fleet operations support. The environmental affairs, chemistry and radiological services manager is responsible for providing technical support in matters related to environmental issues. Some specific responsibilities of the environmental affairs, chemistry and radiological services manager include managing environmental issues such as radiological environmental, nonradiological environmental, dose and shielding calculations, and low-level radioactive waste. The environmental affairs, chemistry and radiological services manager is responsible for overall governance, oversight, and support of plant chemistry, radiation protection, and related functions. The environmental affairs, chemistry and radiological services manager provides direction to plant chemistry and health physics departments regarding operating practices and standards, evaluates programs for conformance to industry best practices, and drives performance improvements where needed.

13.1.1.2.1.2.3.6 Fleet Security Director. The fleet security director is responsible for overall governance, oversight, and support of security-related functions. The fleet security director establishes policy level guidance, provides direction to nuclear plant security departments regarding practices and standards, evaluates programs for conformance to industry best practices, and drives improvements where needed.

13.1.1.2.1.3 Vice President-Vogtle. The vice president-Vogtle reports to the executive vice president/CNO regarding operation issues and support matters. As vice president-Vogtle, he/she is responsible for operation, maintenance, and technical support of VEGP. He/she directs the plant manager, the regulatory affairs manager, the engineering director, the site training manager, site projects manager, and the human resources supervisor.

13.1.1.2.1.4 Vice President-Engineering. The vice president-engineering reports directly to the executive vice president/CNO. This organization includes both project-specific and generic engineering support organizations. Additionally, the vice president-engineering is responsible for nuclear fuel, supply chain, and risk-informed engineering activities. The vice president-engineering directs the risk-informed engineering director, the nuclear fuel director, the fleet engineering director, the projects director, and the supply chain director. Some specific responsibilities include:

A. Design support including creation, revision, and retention of calculations, domestic documents, vendor drawings, and other design basis material.

B. Inservice inspection and testing.

C. Nuclear fuel procurement, nuclear fuel and core design, nuclear fuel reload licensing, nuclear fuel performance, dry storage fuel selection, nuclear fuel procurement vendor oversight.

D. Engineering support on licensing issues.

E. Maintenance of the plant design basis (e.g., calculations, design criteria, and functional system descriptions) for each site.

F. Maintenance of probabilistic risk models and risk-informed applications.

G. Major project management.

H. Procurement activities.

I. Severe accident management activities.

13.1.1.2.1.4.1 Nuclear Fuel Director. The nuclear fuel director reports to the vice president-engineering and directs the core analysis manager, the principal engineer for fuel performance, and the nuclear fuel services manager. The nuclear fuel director is responsible for performance, governance, and oversight of nuclear fuel procurement, nuclear fuel and core design, nuclear fuel reload licensing, nuclear fuel performance, dry cask storage fuel selection, and nuclear fuel vendor oversight for SNC sites.

13.1.1.2.1.4.2 Fleet Engineering Director. The fleet engineering director reports to the vice president-engineering. This individual is responsible for providing corporate governance and oversight to SNC plants in matters related to discipline-oriented design support, configuration management, engineering systems, engineering programs and components, and engineering performance. The fleet engineering director is also responsible for standardization, long-term resource planning, and promoting best practices. Specific responsibilities of engineering services include:

A. Provide expertise to address equipment operability and reliability issues.

B. Provide expertise in the areas of seismic and stress analysis, fire protection, dry spent fuel storage, security, and environmental qualification.

C. Inservice inspection and testing programs.

D. Creation, revision, and retention of calculations, domestic documents, vendor drawings, and other design basis material.

13.1.1.2.1.4.3 Projects Director. The projects director reports to the vice president-engineering and directs the Farley, Hatch, and Vogtle project controls, strategic planning, and project managers. The projects director is responsible for providing governance and oversight for long range project planning and project management services for SNC plants..

13.1.1.2.1.4.4 Risk-Informed Engineering Director. The risk-informed engineering (RIE) director reports to the vice president-engineering and directs the RIE staff. The RIE director is responsible for all RIE activities affecting the plant sites and the corporate office.

13.1.1.2.1.4.5 Supply Chain Director. The supply chain director is responsible for procurement, procurement document control, development of sources of supply including the selection of suppliers to be awarded purchase orders or contracts, and materials management activities.

13.1.1.2.1.5 Vice President and General Counsel. The vice president and general counsel reports to the president/CEO. This individual is responsible for the legal, compliance, and external affairs associated with operation of SNC plants. This individual is also responsible for external affairs activities which include governmental affairs and corporate communications. The vice president and general counsel is also the corporate secretary and directs the managing attorney/compliance manager and the public affairs manager.

13.1.1.2.1.6 Vice President-Regulatory Affairs. The vice president-regulatory affairs reports to the president/CEO. This individual is responsible for licensing through: providing organizational support and management oversight of the sites to assure prompt and proper disposition of regulatory issues; the development of regulatory positions; advising senior management on priorities and activities affecting regulatory at the nuclear sites; and interfacing with NRC management on behalf of the sites. Other responsibilities include: developing policies, standardized processes, and procedures for the maintenance of the licensing basis; the preparation of submittals to the NRC and other regulatory organizations; and the dissemination of regulatory information. Reporting to the vice president-regulatory affairs are the fleet emergency preparedness manager, the fleet performance improvement manager, the regulatory affairs director-fleet, and the regulatory affairs director-nuclear development. The regulatory affairs director-nuclear development is functionally independent of SNC's operating fleet and is noted here for completeness only. Accordingly, the vice president-regulatory affairs is responsible for administration of the corrective action program in the corporate headquarters, the overall coordination of the corporate emergency preparedness programs (including the common Emergency Operations Facility), Emergency Plans, and site emergency response communication.

13.1.1.2.1.6.1 Regulatory Affairs Director-Fleet. The regulatory affairs director-fleet reports to the vice president-regulatory affairs and is responsible for fleet regulatory affairs overall governance, oversight, and support for fleet. This position directs the regulatory affairs programs manager and the licensing manager. The regulatory affairs programs manager and the licensing manager provide matrixed accountability to the vice presidents-plant for each site and are responsible for providing corporate support in matters related to nuclear licensing.

Regulatory affairs-fleet performs both plant-specific and generic nuclear licensing activities. A partial list of responsibilities include:

A. Act as the primary interface with the NRC office of Nuclear Reactor Regulation (NRR) on behalf of the sites.

B. Evaluate regulatory information and interpret NRC requirements.

C. Maintain the Final Safety Analysis Report (FSAR), Technical Specifications, Emergency Plan, Security Plan, and certain other licensing documents for each plant.

13.1.1.2.1.6.2 Fleet Emergency Preparedness Manager. The fleet emergency preparedness manager reports to the vice president-regulatory affairs and is responsible for overall governance, oversight, and support of fleet emergency preparedness activities and programs. The fleet emergency preparedness manager establishes policy level guidance, provides direction regarding practices and standards, evaluates programs for conformance to industry best practices, and drives performance improvements where needed in coordination of the emergency preparedness program and the Emergency Plan.

13.1.1.2.1.6.3 Fleet Performance Improvement Manager. The fleet performance improvement manager reports to the vice president-regulatory affairs and is responsible for fleet performance improvement and for overall governance, oversight, and support for fleet performance improvement activities. The fleet performance improvement manager establishes policy level guidance, provides direction to nuclear plant performance improvement staffs regarding practices and standards, evaluates programs for conformance to industry best practices, and drives performance improvements where needed. The fleet performance improvement manager is also responsible for administration of the corrective action program, fleet procedures program, and documentation management in the corporate headquarters.

13.1.1.2.1.6.4 Executive Vice President Operational Readiness & Site Integration. This executive vice president is responsible for supporting the construction and testing of Vogtle 3-4 and preparing the organization for the operational phase of the project. The executive vice president operational readiness and site integration directs the Vogtle 3-4 operations readiness vice president and the site integration director.

13.1.1.2.1.6.5 Site Integration Director. The site integration director reports to the executive vice president operational readiness & site integration. The site integration director provides direction and oversite of the Vogtle site integration to ensure all regulatory and license issues are resolved. In the initial phase of site integration, the site integration director directs the Vogtle 3-4 security/emergency preparedness project manager, Vogtle 3-4 procedures supervisor, Vogtle 1-2 security manager, and the Vogtle 1-2 emergency preparedness supervisor.

13.1.1.2.1.7 Human Resources Director. The human resources (HR) director reports to the president/CEO. This individual is responsible for the delivery of HR services to the nuclear operating company. This includes HR consulting, organizational effectiveness, succession planning, workforce development, wellness, medical services, fitness for duty, case management, fleet access, and nonunion employee relations. The HR director directs the medical and fitness for duty manager.

13.1.1.2.1.7.1 Medical and Fitness for Duty (FFD) Manager. The medical and FFD manager reports to the Human Resources director and is responsible for coordinating the overall FFD program among SNC management, managing medical services, disability responsibilities, access authorization, the corporate safety and health staff, and the safety and health staff at each of the SNC nuclear plants.

13.1.1.3 Qualifications

Together, SNC and Georgia Power Company operate electric generating plants with an aggregate capacity in excess of 14,000 MWe. SNC personnel have experience in the designing, constructing, startup testing, operating, and staffing of nuclear generating facilities, including HNP, a nuclear power plant with two boiling water reactors and Joseph M. Farley Nuclear Plant, a nuclear plant with two pressurized water reactors.

The corporate organization, which provides the line responsibility for the operation of the VEGP, and the support organization are shown in figure 13.1.1-1. The ultimate responsibility for design, procurement, construction, testing, quality assurance, and operation of the VEGP rests with the president and CEO, who assigns responsibilities to the various organizations as described in paragraph 13.1.1.2.

Members of corporate headquarters staff available for the technical support of the VEGP possess that combination of education, experience, and skills commensurate with their level of responsibility. They provide reasonable assurance that support services provided in connection with the operation of the VEGP units will not constitute a hazard to the health and safety of the public.

The operating organization for the VEGP is described in subsection 13.1.2. The company technical support organizations for operation, modification, and maintenance are described in paragraph 13.1.1.2 and shown on figure 13.1.1-1. The operating and support organization described herein provide assurance of safe operation of VEGP Units 1 and 2 in compliance with regulatory requirements.

13.1.2 OPERATING ORGANIZATION

13.1.2.1 Plant Organization

The VEGP consists of two nearly identical nuclear generating units. The applicable plant organization is shown in figure 13.1.2-1.

13.1.2.2 Plant Personnel Responsibilities and Authorities

13.1.2.2.1 Overall Plant Management

The executive vice president operational readiness & site integration is responsible for supporting the construction and testing of Vogtle 3-4 and preparing the organization for the operational phase of the project. The executive vice president operational readiness and site integration directs the Vogtle 3-4 operations readiness vice president and the Vogtle site integration director.

The site integration director reports to the executive vice president operational readiness & site integration. The site integration director directs the Vogtle 3-4 security/emergency preparedness project manager, Vogtle 3-4 procedures supervisor, Vogtle 1-2 security manager, and the Vogtle 1-2 emergency preparedness supervisor and provides direction and oversite of the Vogtle site integration.

The Vogtle 3-4 security/emergency preparedness project manager reports to the site integration director. This leadership position is responsible for developing the necessary plans, schedules, processes, procedures, and budgets to effectively prepare Vogtle 3-4 to integrate with Vogtle 1-2. The Vogtle 3-4 security project manager coordinates and collaborates with Vogtle 1-2 security to develop a single security plan for the Vogtle 1-2-3-4 site. Reporting to the Vogtle 3-4 security/emergency preparedness project manager is the 3-4 security supervisor.

Reporting to the Vogtle 3-4 security/emergency preparedness project manager is the 3-4 security supervisor.

The vice president-Vogtle is responsible for direct management of the plant, including industrial relations, planning, coordination, direction of operation, training, maintenance, refueling, and technical activities. The vice president-Vogtle is responsible for:

- Compliance with the requirements of the operating license, Technical Specifications, and quality assurance program, and
- Approval, prior to implementation, of each proposed test, experiment, or modification to systems or equipment that affects nuclear safety.

In the vice president-Vogtle's absence, the plant manager assumes responsibility for the plant. The vice president-Vogtle will designate in writing other qualified personnel to assume overall plant responsibility in his absence. (See paragraph 13.1.2.2.2 for succession of responsibility for overall plant operation.)

The vice president-Vogtle reports to the executive vice president/CNO. The vice president-Vogtle has access to the advice and services of technical specialists in the corporate office and outside expertise as necessary.

The positions reporting to the vice president-Vogtle are:

A. The plant manager is responsible for the operation of Units 1 and 2. Reporting to the plant manager are the operations director, the maintenance director, the work management director, the health physics manager, and the chemistry manager.

B. The regulatory affairs manager is responsible for plant procedures, site licensing, and site performance improvement. Reporting to the regulatory affairs manager are the plant procedures supervisor, the site licensing supervisor, and the performance improvement supervisor.

C. The engineering director is responsible for providing engineering support for Units 1 and 2. Reporting to the engineering director are the design engineering manager, the engineering systems manager, the engineering programs manager, and the reactor engineering supervisor.

D. The site projects manager is responsible for providing oversight of project activities by ensuring outage activities are integrated and supported, budgets are maintained, and schedules and objectives are met including projects related to severe accident management.

E. The human resources supervisor is responsible for human resources support activities.

F. The training manager is responsible for the training and retraining programs and the training simulator.

Responsibilities of the individuals reporting to the plant manager are as follows:

A. The operations director is responsible for operating the plant safely and efficiently.

B. The maintenance director is responsible for directing the maintenance teams, for performance of preventive maintenance, for repair of plant equipment, for assigned quality control measures, and for providing maintenance support utilizing contractor craft personnel.

C. The health physics manager and the chemistry manager are responsible for the radiation protection program and for chemistry and radiochemical activities, respectively, at VEGP.

D. The work management director is responsible for planning, scheduling, and coordinating outages and for developing daily work plans which include technical specification surveillances, corrective maintenance, and planned maintenance.

Responsibilities of the individuals reporting to the regulatory affairs manager are as follows:

A. The site licensing supervisor is responsible for site licensing activities.

B. The procedures supervisor is responsible for site procedures.

C. The performance improvement supervisor is involved in day-to-day safety-related activities and seeks to find improvement in plant processes and actions. The performance improvement supervisor is also responsible for the corrective action program.

Responsibilities of the individuals reporting to the engineering director are as follows:

A. The engineering systems manager is responsible for systems engineering and equipment reliability. Additional duties include providing direction to other plant departments in technical matters and plant design.

B. The design manager is responsible for preparation of design changes and implementation of design changes, implementation of the quality control inspection program, configuration management, and the rapid response team. Additional duties include providing direction to other plant departments in technical matters and plant design.

C. The engineering programs manager is responsible for engineering programs, components, and predictive maintenance. Additional duties include providing direction to other plant departments in technical matters and plant design.

D. The reactor engineering supervisor is responsible for reactor physics and providing direction to other plant departments in technical matters.

13.1.2.2.2 Operations Supervision and Shift Organization

The operations director is responsible to plant management for the operation of Unit 1 and Unit 2. The operations director or at least one operations superintendent shall possess a senior reactor operator (SRO) license. Reporting to the operations director are the operations superintendents, operation support superintendent, and the shift managers. The operations superintendent (daily) directs day-to-day planning and provides technical support; the operations support superintendent supervises staff support functions and coordinates training of

operations personnel. The operations superintendent (outage) prepares for and conducts outage related activities and oversees outage work activities, surveillances, and tests.

The shift manager is responsible for seeing that plant operations are conducted in accordance with appropriate standing orders, unit operating procedures, and technical specifications. The shift manager's principal responsibility is ensuring safe operation during his assigned shift as addressed in the requirements of item I.A.1.2 of NUREG-0737. The shift manager shall possess an SRO license.

The shift technical advisor position meets the intent of NUREG-0660, as clarified by NUREG-0737, item I.A.1.1. The shift technical advisor position may be eliminated if the qualifications of the shift manager, shift supervisor, or SRO-licensed shift support supervisor meet the requirements of the shift technical advisor position. Section 13.2 describes shift technical advisor training, and subsection 13.1.3 describes shift technical advisor qualifications.

A shift supervisor for each unit is under the supervision of the shift manager. Each shift supervisor shall possess a SRO license. Each shift supervisor is responsible for the safe and efficient operation of the unit. The shift supervisor for each unit keeps a record of shift activities and establishes unit load as requested by the system operator or as emergency conditions dictate.

Also reporting to the shift supervisors are the shift support supervisors, plant operators, and systems operators.

Plant operators monitor the plant status and operate equipment as needed to maintain control of the various plant processes. Most of their duties are located in the control room; although, they may perform inspections in other areas of the plant. The operating crew may make radiation and contamination surveys within the plant. (In addition to the control room personnel, a chemistry technician is on duty during plant operations.) The Technical Specifications state the shift manning requirements for all modes of operation.

The succession to responsibility for overall plant operation and the authority to issue operating instructions or special orders, in the event of absences, incapacitation of personnel, or other emergencies, are as follows:

A. Plant manager (PM).

B. Operations director.

C. Regulatory affairs manager.

D. Licensed manager/superintendent designated by PM.

E. Shift manager (SRO).

F. Shift supervisor (SRO).

13.1.2.2.3 Work Management, Health Physics, Chemistry, and Maintenance Supervision

The work management director reports to the plant manager and supervises the work management staff. The function of the work management staff is to monitor and enhance plant performance and provide maintenance support.

The health physics manager reports to the plant manager and is responsible for the radiation protection program. The health physics manager also ensures that appropriate monitoring devices and protective clothing are available and is responsible for radiation monitoring devices used by personnel entering the plant and for the maintenance of all required radiation exposure

records of plant support and visiting personnel. Reporting to the health physics manager is the health physics staff.

The chemistry manager reports to the plant manager and is responsible for chemical and radiochemical activities at the plant. The chemistry manager is responsible for administrative control of effluent releases from the plant to ensure that the releases are maintained ALARA and within the required limits.

The chemistry support supervisor and plant chemist work under the direction of the chemistry manager and are responsible for performing chemical and radiochemical sample analyses, and for monitoring and maintaining the plant makeup water demineralizer to ensure proper water quality. Laboratory personnel advise operations staff on the operation of plant systems to maintain water quality within the specified limits.

The maintenance director directs and plans maintenance activities with the assistance of other departments. The maintenance director is also responsible for ensuring the building and grounds upkeep. Reporting to him are maintenance superintendents and team leaders. The team leaders and assistant team leaders direct the team technical staff, electricians, I&C technicians, and mechanics in the upkeep of equipment. The team leaders and assistant team leaders are responsible for directing technicians in testing, calibration, surveillance checks, and repair of plant instrumentation and control systems. A maintenance superintendent reporting to the maintenance director also supervises a group of plant helpers who conduct minor maintenance, painting, and decontamination activities. The maintenance director oversees outage craftsmen who are brought in during outages to supplement the permanent staff.

13.1.2.2.4 Quality Assurance and Quality Supervision

The vice president-Vogtle is responsible for the development and implementation of the quality assurance program to comply with the SNC QATR with the exception of the controls assigned to the quality assurance (QA) organization. Reporting to the engineering systems manager, the designated engineering supervisor will provide for inspection activities required by the quality assurance program. Personnel performing quality control inspections have procedural authority to stop work and to control further processing, use, or installation of nonconforming items.

The nuclear oversight manager is the senior manager responsible for quality assurance described in the SNC QATR and is responsible for managing activities of the nuclear oversight organization. The nuclear oversight organization will provide a comprehensive independent audit of safety-related activities to verify that they are in compliance with the quality assurance program. The quality assurance program during operations is discussed in the QATR.

13.1.2.3 Operating Shift Crews

A total manpower in excess of five full shift crews is provided to avoid excessive overtime.

During refueling operations, when the reactor core configuration is being altered, a senior reactor operator will supervise the fuel handling activities in the reactor containment.

Licensed operating personnel are trained in the radiation protection procedures and are capable of performing routine or special radiation surveys using portable radiation detectors. These operators are trained in the use of protective barriers and signs, protective clothing and breathing apparatus, in the performance of contamination surveys, and in performance of checks on radiation monitors. Plant personnel will be trained in the limits of exposure rates and

accumulated dose. The shift manager is responsible for implementing the radiation protection program in the absence of the health physics manager or his designated alternate.

13.1.3 QUALIFICATIONS OF PERSONNEL

13.1.3.1 Qualification Requirements

Each member of the unit staff shall meet or exceed the minimum education and experience recommendations of Regulatory Guide 1.8, Revision 2, and for licensed staff, 10 CFR 55.59. Prior to meeting the recommendations of Regulatory Guide 1.8, Revision 2, personnel may be trained to perform specific tasks and will be qualified to perform those tasks independently. Personnel who complete an accredited program which is endorsed by the Nuclear Regulatory Commission (NRC) shall meet the requirements of the accredited program in lieu of the above.

Table 13.1.3-1 shows the education and experience requirements for VEGP organizational titles with equivalent titles taken from ANSI 18.1-1971 and ANSI 3.1-1981 for those positions, as endorsed by Regulatory Guide 1.8, Revision 2, for key plant personnel. Personnel in job positions covered by accredited training programs meet the requirements of those accredited programs as described in section 13.2.

13.1.3.2 ANSI 18.1-1971 Definitions

Nuclear power plant - A nuclear power plant is any plant using a nuclear reactor to produce electric power, process steam, or space heating.

Experience - Experience shall mean actual applicable working experience in design, construction, startup, operation, maintenance, or technical services. Observation of others performing applicable functions shall not be considered acceptable experience.

Academic training - Academic training is successfully completed college-level work leading to a recognized degree.

Related technical training - Related technical training is formal training beyond the high school level in technical subjects associated with the position in question, acquired in training schools or programs conducted by the military, industry, utilities, universities, vocational schools, or others. Such training programs should be of a scheduled and planned length and include text material, lectures, and frequent examinations.

On-the-job training - On-the-job training is participation in nuclear power plant startup, operation, maintenance, or technical services under the direction of appropriately experienced personnel.

13.1.3.3 Qualification of Plant Personnel Using ANSI 18.1-1971

Nuclear power plant experience is nuclear experience acquired in the design, construction, startup, or operation of nuclear power plants. Further, nuclear experience acquired at military, nonstationary, or propulsion nuclear plants may qualify as equivalent experience on a one-for-one time basis.

Nuclear experience acquired in nonpower plants (such as test, research, or production reactors) may qualify as equivalent to nuclear power plant experience on a one-for-one time basis. Only 1 year of such experience will be accredited toward the total nuclear experience qualification.

Training may qualify as experience if acquired in appropriate reactor simulator training programs, on the basis of 1 month's training being equivalent to 3 month's experience. Only one year of such experience will be accredited toward the total nuclear experience qualification.

Training programs, the culmination of which involves actual reactor operation, may qualify as equivalent to nuclear power plant experience on a one-for-one time basis for up to a maximum of 1 year credit.

On-the-job training may qualify as equivalent to nuclear power plant experience on a one-for-one time basis for up to a maximum of 1 year credit toward the nuclear power plant experience requirement.

The qualification of the staff personnel holding key managerial and supervisory positions as described in paragraph 13.1.2.2.1 are provided in table 13.1.3-1.

For qualification of personnel for positions that require a senior reactor operator or reactor operator license, the shift technical advisor and the radiation protection manager will follow the recommendations of ANSI 3.1-1981 as modified by Regulatory Guide 1.8, Revision 2.

TABLE 13.1.3-1 (SHEET 1 OF 3)

ACCEPTABLE EDUCATION AND EXPERIENCE FOR SELECTION AND TRAINING OF VEGP PERSONNEL

		Recommended Experience (years)						Suggested Education (years)			
						License					
ANSI N18.1 Title	VEGP Title	Total Power Plant	Nuclear Power Plant	Other Applicable	Academic Training	Reactor Operator	Senior Reactor Operator	Academic	Related Technical Training	Amount of Education Creditable for Experience	Remarks
Managers											
Plant manager	Executive Vice President Operational Readiness & Site Integration	10	3				X[a]	4		4	
	Vice President-Vogtle	10	3				X[a]	4		4	
Assistant plant manager	Regulatory Affairs managers	10	3				X[a]	4		4	
Operation manager	Operations director	8	3				X[d]	2 or	2	2	
Technical manager	Engineering director[e]/Design/ Engineering support managers/Technical Services manager		1	7				4		4	
Maintenance manager	Maintenance director	7	1					2 or	2	2	
Training manager (b)	Training manager		2	4				4			
Superintendents or supervisors not requiring NRC license											
Radiochemistry	Chemistry manager or plant chemist			5				2 or	2	4	1 year experience in radiochemistry.
	Team leader			7							Knowledge of area of responsibility and high school diploma or equivalent required.

TABLE 13.1.3-1 (SHEET 2 OF 3)

		Recommended Experience (years)						Suggested Education (years)			
						License					
ANSI N18.1 Title	VEGP Title	Total Power Plant	Nuclear Power Plant	Other Applicable	Academic Training	Reactor Operator	Senior Reactor Operator	Academic	Related Technical Training	Amount of Education Creditable for Experience	Remarks
	Team leader			7							Knowledge of area of responsibility and high school diploma or equivalent required.
Foreman	HP or chemistry foreman			4							High school diploma or equivalent required.
I&C assistant team leaders				5				2 or	2	4	6 months experience in nuclear I&C.
Assistant team leader				4							Knowledge of area of responsibility and high school diploma or equivalent required.
Technical support personnel											
Engineer-in-charge	Engineering director[e]		3		4						
ANSI 3.1 Title											
Radiation protection manager	Health physics manager, plant health physicist, or HP support supervisor		3[c]	4				4			

TABLE 13.1.3-1 (SHEET 3 OF 3)

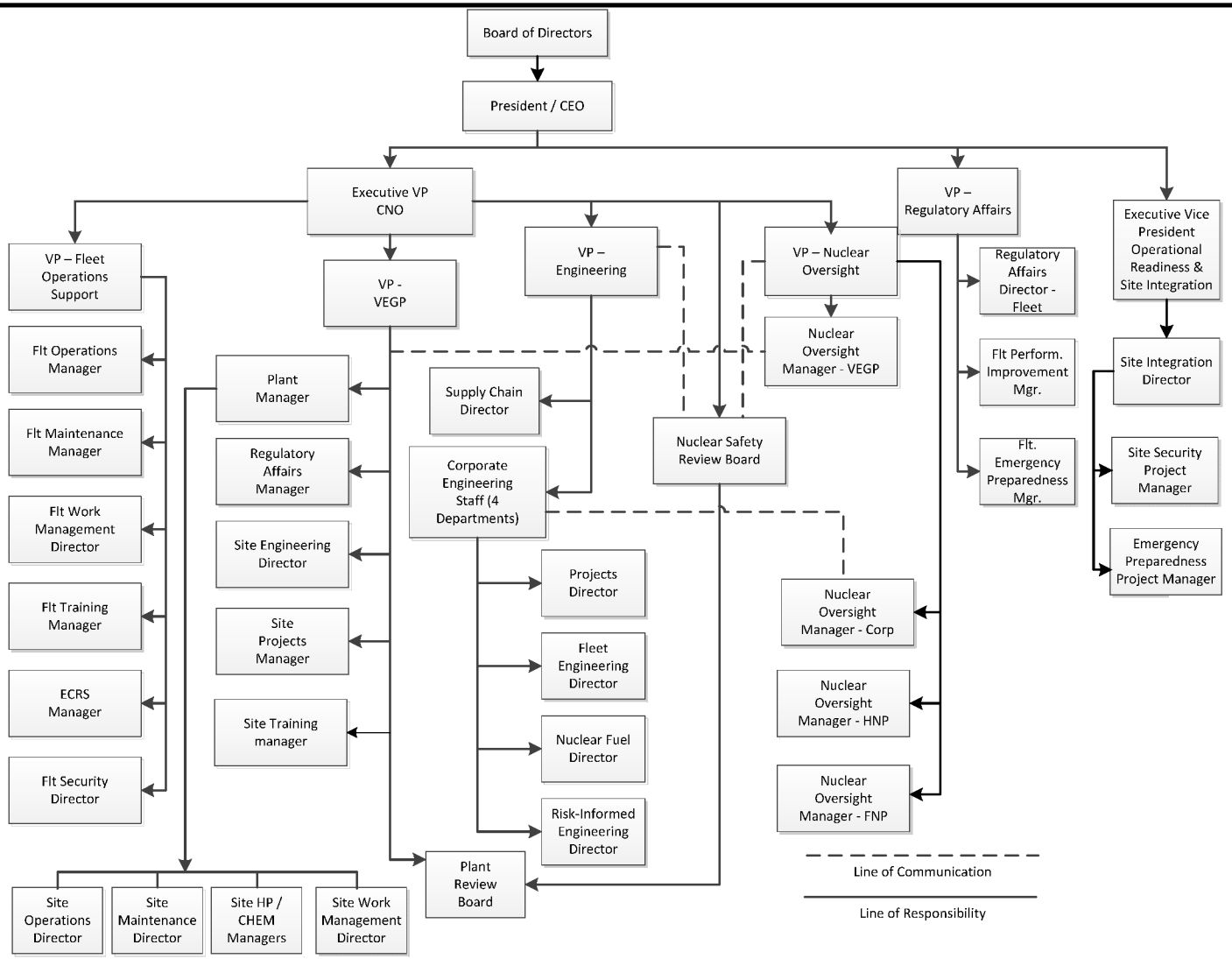
a. Senior reactor operator license is not required for the vice president-Vogtle, executive vice president operational readiness & site integration, the plant manager, or the regulatory affairs manager, but the vice president-Vogtle or the plant manager will have (or have held) an SRO license or have the background required to sit for examination.

b. Required by NUREG-0737.

c. Either the health physics manager, the health physics support supervisor, or the plant health physicist will be assigned the radiation protection manager responsibilities. In addition to the above, the selected individual will have participated in radiation protection activities during at least 2 months of operation at greater than 20 percent power, a routine (1 to 2 months) refueling outage, and will have at least 6 months of nuclear experience at VEGP.

d. The operations director or at least one operations superintendent shall hold an SRO license.

e. The engineering director fulfills both the technical manager and engineer-in-charge functions; therefore, the engineering director must meet both sets of requirements.



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SOUTHERN COMPANY

VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2

NUCLEAR OPERATIONS ORGANIZATION
VEGP UNITS 1 AND 2

FIGURE 13.1.1-1

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SOUTHERN COMPANY

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

NUCLEAR OPERATIONS ORGANIZATION
VEGP UNITS 1 AND 2

FIGURE 13.1.2–1

13.2 TRAINING PROGRAMS

Personnel meet the qualification and training recommendations of Regulatory Guide 1.8, Revision 2, and for the licensed staff, 10 CFR 55.59, before they are considered qualified to perform all duties independently. Prior to meeting the recommendations of Regulatory Guide 1.8, Revision 2, personnel are trained to perform specific tasks and are qualified to perform those tasks independently. Personnel who complete an accredited program endorsed by the Nuclear Regulatory Commission (NRC) meet the requirements of the accredited program in lieu of Regulatory Guide 1.8, Revision 2.

13.2.1 ACCREDITED TRAINING PROGRAMS

The VEGP training programs have been developed in accordance with the systems approach to training as described by the Institute for Nuclear Power Operations (INPO). The National Academy for Nuclear Training, through a formal accreditation process, verifies that VEGP training programs meet the established criteria. VEGP is a branch of the National Academy and has achieved accreditation of the following programs:

- Nonlicensed operator
- Reactor operator
- Senior reactor operator
- Continuing training for licensed personnel
- Shift supervisor
- Shift technical advisor
- Instrument and control (I&C) technician
- Electrical maintenance personnel
- Mechanical maintenance personnel and supervisor
- Radiological protection technician
- Chemistry technician
- Engineering support personnel

The training programs are periodically evaluated and revised as appropriate, and reviewed by management for effectiveness.

13.2.1.1 Training Program Records

Records to document course completion and documentation of qualification are retained as quality records in accordance with the SNC Quality Assurance Topical Report (QATR).

13.2.1.2 Simulator

The VEGP simulator conforms to the guidance given in Regulatory Guide 1.149, Revision 1. The certification of the VEGP simulator as a plant-referenced simulator is complete and certification shall be maintained as described in 10 CFR 55.45 (b)(5). Exceptions to ANSI 3.5 are listed in the certification report filed with the NRC every 4 years.

13.2.1.3 Instructor Qualification

Accreditation of training programs includes criteria for qualification and requalification of instructors. In addition, the fire protection training instructor also meets the requirements outlined in paragraph 9B.C.3.d(2).

13.2.2 OTHER TRAINING PROGRAMS

13.2.2.1 General Employee Training Program

All persons regularly employed at VEGP are trained in the following areas commensurate with their job duties:

- Fitness for duty
- General plant description
- Radiological protection
- Emergency preparedness
- Industrial safety
- Fire protection
- Security
- Quality assurance

13.2.2.2 Fire Brigade Training

Personnel assigned to any fire brigade complete initial training and periodic retraining as designated by appendix 9B.

13.2.2.3 Quality Control Training Program

Plant personnel who perform quality control (QC) inspections, both receipt and in-plant, shall meet the minimum education and experience requirements of ASME NQA-1-1994 as described in the SNC QATR prior to being certified to perform inspections. Initial training consists of quality assurance procedures training, codes and standards training, and on-the-job training.

On-the-job training for receipt personnel and QC inspection personnel is based on job requirements. Initial training has a minimum duration of 40 h. QC inspection personnel and the engineering supervisor (programs) responsible for inspection activities must receive additional training in plant systems. QC inspection personnel will complete the plant systems training within 1 year of assignment to their position and the engineering supervisor must complete this training prior to being qualified. Material receipt inspection personnel are not required to receive additional training in plant systems.

After completing initial qualification, QC plant inspection and material receipt inspection personnel normally receive training annually to improve and maintain their inspection skills. This annual training will be in subjects applicable to their inspection duties and qualification.

13.2.2.4 Mitigating Core Damage Training Program

The VEGP training program for mitigating core damage is not a separate program for licensed personnel, but is integrated into licensed personnel training, senior reactor operator certified personnel training, and shift technical advisor training. Other personnel, including the health physics manager, chemistry manager, and team leaders and assistant team leaders for I&C technicians in the maintenance department, as well as the plant manager and the operations manager, complete training in mitigating core damage commensurate with their responsibilities.

13.3 EMERGENCY PLANNING

A comprehensive emergency plan for VEGP Units 1 and 2 is provided as a separate volume.

13.4 OPERATIONAL REVIEW

Operating activities that affect nuclear safety are reviewed. The review program was implemented prior to initial fuel loading and ensures review and evaluation of tests and experiments, unplanned events, and proposed changes. The program complies with the requirements of 10 CFR 50.59 relating to proposed changes, tests, and experiments. This program is conducted in accordance with the requirements of the SNC Quality Assurance Topical Report (QATR).

The vice president-Vogtle has responsibility for safe operation of the plant. He is kept abreast of plant operating conditions by the managers, superintendents, and supervisors who are knowledgeable of and experienced in their areas of job responsibility. These individuals monitor operating and maintenance activities as part of their normal duties.

In addition, a formal review program is carried out for changes to systems, procedures, tests, experiments, and after-the-fact review and evaluation of unplanned events that affect nuclear safety. The vice president-Vogtle shall approve, prior to implementation, each proposed test, experiment, or modification to systems or equipment that affect nuclear safety. In addition to this requirement for vice president-Vogtle approval as described above, this program is also implemented through standing committees, as referenced below.

13.4.1 ONSITE REVIEW

Refer to the SNC QATR for discussion related to the Plant Review Board (PRB), which performs the onsite review function.

13.4.2 INDEPENDENT REVIEW

Refer to the SNC QATR for discussion related to the Nuclear Safety Review Board (NSRB), which performs the independent review function.

13.5 PLANT PROCEDURES

This section describes administrative, maintenance, and operating procedures that are used by the operating organization to ensure that routine, off-normal, and emergency activities are conducted in a safe manner. Operations affecting safety are conducted in accordance with detailed written and approved procedures.

13.5.1 ADMINISTRATIVE PROCEDURES

Administrative procedures provide rules, instructions, policies, practices, or guidelines for the plant staff. VEGP administrative procedures establish controls which govern changes to programs and changes or modifications to systems or equipment affecting nuclear safety. These controls ensure that these programs and changes or modifications are independently reviewed by qualified personnel and approved by appropriate personnel.

13.5.1.1 Administrative Procedure Descriptions

A. Procedures for Preparation, Review, and Control of Procedures

These procedures establish the controls for the preparation, review, and control of plant procedures which cover activities described in the SNC Quality Assurance Topical Report (QATR).

Included within these procedures are provisions to ensure that new or revised procedures are reviewed for adequacy by appropriately qualified personnel other than the originator. In accordance with these procedures, a reviewer shall acquire knowledge of the applicable requirements for a procedure and shall verify the proper incorporation of these requirements in the procedure; additionally, a reviewer shall assess whether the procedure or procedure revision potentially involves a change in the VEGP Technical Specifications or requires prior NRC approval pursuant to 10 CFR 50.59. The acceptability of procedure reviews is assured by the review and approval by the responsible VEGP department heads or their designee. Department heads shall further ensure that procedures requiring Plant Review Board (PRB) review in accordance with the SNC QATR are forwarded to the PRB.

For procedures not forwarded to the PRB, reviewers will meet requirements of section 4.4 of ANSI N18.1-1971 for applicable disciplines. For those disciplines not described in section 4.4 of ANSI 18.1-1971, the reviewer will have a minimum of 5 years experience. A maximum of 4 years of this 5 years may be fulfilled by related technical or academic training. Reviewers of quality control inspection procedures shall meet the requirements of ASME NQA-1-1994, as described in the QATR. Also, those procedures not forwarded to the PRB and impacting another department's area of responsibility, shall be forwarded to the impacted departments for their review.

The vice president-Vogtle has ultimate responsibility for all plant procedures. Specific procedures for which vice president-Vogtle approval is required are discussed in plant procedures. Department heads are established as the approving authority for other procedures covering activities within their area of

responsibility. Additional guidance regarding this approval authority for specific procedures is provided in plant procedures.

Additional provisions of these procedures exist to ensure that changes or revisions to procedures are reviewed and approved in accordance with the same administrative controls used for review and approval of new procedures. A provision is made to ensure that EOPs and AOPs are reviewed at least every 2 years by a knowledgeable individual to determine whether changes are necessary or desirable.

As a part of the overall quality assurance program, the QA group performs various audits (described in the QATR) to assure that the procedural process is working and that procedures are being properly maintained. Also, provisions exist to ensure that procedures, once approved, are distributed appropriately so that only the most current procedures are used by plant personnel.

B. Procedures for Making Temporary Changes to Procedures

This procedure provides the method for making a temporary change to an approved plant procedure.

Included in this procedure is a provision to ensure that any such change be approved by at least two knowledgeable members of the plant staff, one of whom shall hold a senior operator license. Also included in this procedure is a provision to ensure that temporary changes to procedures are documented and incorporated as appropriate into the next revision of the procedure.

Additional guidance regarding temporary changes to procedures for specific procedures is provided in the QATR.

C. Procedures for Feedback of Operating Experience

These procedures include measures to ensure that pertinent operating experience information originating from both within and outside of the plant organization is fed back to operators and other appropriate personnel in accordance with NUREG-0737, item I.C.5. This information can include industry event reports, vendor reports, inhouse event reports, and Nuclear Regulatory Commission publications. These procedures establish management responsibility for direction, control, and administration of the operations assessment program. Included in these procedures is the identification of organizational responsibilities for reviewing and prioritizing operating experience, and for ensuring distribution of pertinent information to the appropriate plant personnel. Steps exist to ensure that information is reviewed by individuals of appropriate technical knowledge and that appropriate corrective actions (such as procedure or program revisions), if needed, are specified. The procedures include steps which ensure records are maintained to reflect the status and disposition of operating experience evaluations and their associated corrective actions. Completion status and followup verification status of completed action is tracked to ensure implementation. Additional steps ensure that plant personnel do not routinely receive a large volume of operating experience that might obscure the lessons to be learned from more significant events, and also that the program receives periodic evaluation for effectiveness.

D. Procedures for Control Room Access

These procedures give the shift supervisor authority to limit access to the control room to those individuals responsible for the direct operation or support of the

plant. Included are steps that direct personnel other than the onshift operations crew and the operations chain of command to request permission of the shift operations crew to enter the limited access area shown in figure 13.5.1-1. Also, these procedures establish good conduct rules for personnel within the control room area to avoid any disruption of operating activity. Responsibility for limiting control room access is a normal shift duty of the shift supervisor. The shift supervisor is additionally given the authority to require that any individual granted access to the control room immediately leave the control room if that individual interferes with the duties of the control room operators in maintaining safe operation of the plant, or exhibits behavior contrary to good conduct rules. These procedures address the requirements of NUREG-0737, item I.C.4, for control room access; while the requirements for establishing the lines of authority, responsibility, and succession in the control room are addressed by the procedures described in section E.

E. Procedures for Operating Duties, Responsibilities, and Authority

These procedures clearly describe the duties, responsibilities, and authority for the control room personnel, which include the shift manager, the shift supervisor, the shift support supervisor, the systems operator, the plant operator, and the shift technical advisor. The command line of authority for these personnel is established by these procedures. Specifically, the shift manager is established as the senior operations representative on each shift and shall have responsibility for the safe and efficient operation of the plant. The shift supervisor is established as having responsibility for the safe and efficient operation of his assigned unit, since a shift supervisor is assigned to each operating unit on each shift. Included in these procedures are provisions for the shift manager to maintain a broad perspective of operational conditions affecting the safety of the plant at all times and provisions for him not to become totally involved in any single operation during plant transients or emergency conditions. Other provisions establish the shift manager as having the authority and responsibility to declare emergencies, and to function as the emergency director, until being relieved of this responsibility by a higher-ranking qualified manager.

Provisions for ensuring that the shift manager is not routinely performing administrative functions that could detract from or that are subordinate to his command function and his responsibility for ensuring the safe operation of the plant are exemplified by the following:

1. An operations individual who is not on shift shall prepare work and vacation schedules.
2. The shift supervisor shall issue clearances for equipment within his assigned unit.
3. The shift supervisor shall have the administrative duty of limiting access to the control room.

The plant manager annually reviews the administrative duties of the shift manager to ensure that he is not routinely performing administrative duties as described above.

The procedures described in this section are in accordance with requirements of NUREG-0737, item I.A.1.2, item I.C.3, and item I.C.4. The process for handling standing orders is addressed in section F.

F. Procedures for Standing Orders

These procedures provide for the issue of temporary instructions to plant operating personnel to address subjects not covered by existing plant operating procedures. Included in these procedures are provisions for maintaining and implementing approved standing orders, and periodically reviewing standing orders for continued applicability.

G. Procedures for Shift Manning and Overtime Restrictions

These procedures establish the normal and minimum shift positions that must be manned for operation of the plant. Included is the number of individuals to fill these normal and minimum positions for both one-unit and two-unit operation. These procedures restrict the use of overtime that may be scheduled to meet the shift crew staffing requirements, such that overtime use does not exceed the requirements of 10 CFR Part 26, Subpart I.

H. Procedures for Shift Relief and Turnover

These procedures ensure that a comprehensive exchange of information takes place between the oncoming and offgoing shift personnel so that the oncoming shift is aware of critical plant status information and system availability prior to assuming duty. Included are provisions to ensure that each oncoming licensed operator reviews and signs the checklist for his position prior to assuming his duties. Both licensed and nonlicensed operators review logs applicable to their positions prior to assuming their duties. In addition, nonlicensed operators complete round sheets early in each shift. All oncoming individuals discuss relevant items affecting plant operation with the offgoing individuals prior to assuming their duties. Provisions also include ensuring that an individual is qualified for the position that he will assume. These procedures establish, as part of the offgoing control room operator responsibility, the need to ensure that his relief is fully aware of existing plant conditions, and that his relief is alert, coherent, and fully capable of performing his assigned duties. Provisions are provided to ensure that other appropriate supervisory personnel, including maintenance performance team shift personnel, develop shift logs to update oncoming shifts of plant activities. These provisions also ensure that oncoming shift personnel review and sign shift logs for their position. These procedures are in accordance with the requirements of NUREG-0737, item I.C.2.

I. Equipment Control Procedures

These procedures provide instructions for releasing plant equipment or systems for maintenance, testing, or inspection; they establish the shift supervisor as the responsible authority for issuing and releasing clearances for equipment to be taken out of service within his assigned unit. The provisions of these procedures include steps to ensure that equipment taken out of service and placed in a controlled status is clearly identified by the use of tagging. Other provisions provide for a second qualified person verifying the isolation or restoration of a safety-related component or system, including proper realignment, unless functional testing can be performed to prove the correct realignment of all equipment, valves, and switches involved. A second verification of safety-related system alignment will not be made in cases of significant radiation exposure, or where alignment changes or alterations of status do not render the component/systems incapable of performing designed safety functions. Additional provisions are made to maintain the status of equipment and to

determine the operability of equipment upon return to service; on return to service, a system lineup verification will be made, and additional lineup verifications may be made at periodic intervals while in service. These procedures address the requirements of NUREG-0737, item I.C.6 and item II.K.1.10, and the requirements of ANS N18.7-1976, Section 5.2.6, relative to equipment control.

J. Maintenance and Modification Administrative Control Procedures

Maintenance of equipment and plant modifications important to plant safety are performed in accordance with written procedures as described in paragraph 13.5.2.2. Administrative controls are provided to ensure compliance with applicable codes, regulations, and requirements.

K. Fire Protection Procedures

The VEGP fire protection program is governed and implemented through the use of fire protection procedures. These procedures provide guidelines for the following: administrative controls, system operation, firefighting activities, fire brigade and general personnel training, and agreements with local offsite fire departments, as described in subsection 9.5.1.

L. Crane Operation Procedures

Crane administrative operating procedures require that crane operators who operate cranes over fuel pools be qualified and conduct themselves in accordance with ANSI B30.2-1976 (chapters 2 and 3), Overhead and Gantry Cranes.

M. Temporary Procedures

Temporary procedures are issued, as required, to provide detailed instructions for specific jobs lasting a specific duration.

N. Emergency Core Cooling System Outage Data Collection Procedures

As discussed in Generic Letter 83-37, the NRC has completed their review of ECCS data and determined that no changes in the Technical Specifications were required. Therefore, the surveillance and reporting requirements in the VEGP Technical Specifications adequately address ECCS outages, and this data will no longer be reported to the NRC.

O. Plant Security Procedures

Plant security procedures provide for the implementation of the security plan. (See section 13.6.)

13.5.2 OPERATING AND MAINTENANCE PROCEDURES

13.5.2.1 Operating Procedures

These procedures are for operation of plant equipment. These procedures are developed and provided to ensure an effective system for verifying the correct performance of operating activities and to reduce human error. These procedures meet the intent of NUREG-0737, item I.C.6. Conformance of these procedures with ASME NQA-1-1994 is discussed in the SNC QATR.

Although format can be varied depending on their nature, these procedures usually have the following format:

1.0 PURPOSE

The purpose for which the procedure is intended is to be briefly and clearly stated.

2.0 PRECAUTIONS AND LIMITATIONS

Precautions are to be established to alert the individual performing the task of those important measures which will be used to protect equipment and personnel, or will serve to avoid an abnormal or emergency situation.

Limitations on the parameters being controlled and appropriate corrective measures to return the parameter to normal are established. Control band may be specified where applicable.

3.0 PREREQUISITES OR INITIAL CONDITIONS

Each procedure will identify those independent actions or procedures which must be completed, or those plant conditions which must exist, prior to its use. Prerequisites applicable only to certain sections of a procedure will be so identified.

4.0 MAIN BODY

The main body of a procedure will contain step by step instructions in the degree of detail necessary for a qualified person performing a function or task.

Subsections of this section will vary according to procedure type. Format for each procedure type will be consistent.

5.0 ACCEPTANCE CRITERIA

Procedures will contain, where applicable, acceptance criteria against which the success or failure of test type activity may be judged. This section may be omitted if none are required.

6.0 REFERENCES

References, including reference to Technical Specifications, will be included in procedures as applicable.

These procedures are classified as follows:

A. Plant/Unit Operating Procedures

These are procedures for integrated plant operations, such as mode changes.

B. System Operating Procedures

These procedures are for normal, alternate means of operation and limits of operation of individual plant systems, including removal from service and return to service.

C. Surveillance Procedures (Operations Department Responsibility)

These procedures, which describe how to perform testing functions, are required to verify operational readiness of structures, systems, and components.

D. Annunciator Response Procedures

These procedures specify responses to annunciated alarms. They outline automatic system response and operator actions to mitigate the conditions.

These procedures are indexed by light board number and alarm window coordinates to facilitate operator response. Any annunciator response procedure providing response to an emergency condition reference the appropriate emergency operating procedure.

E. Abnormal Operating Procedures

These procedures specify operator actions for restoring a plant operating variable to its normal controlled value when it departs from its range or operator actions necessary to restore normal operating conditions following a transient. Such actions are invoked following an operator observation or any annunciator alarm indicating a condition which, if not corrected, could degenerate into a condition requiring action under an emergency operating procedure.

F. Emergency Operating Procedures

The emergency operating procedures are developed according to an emergency operating procedure writer's guide.

The emergency operating procedures are developed according to an action plan, which incorporates developmental information provided in the implementation plan guideline published by the Institute of Nuclear Power Operations.

Technical guidelines (emergency response guidelines) and accompanying background/support information are supplied by the Westinghouse Owners Group. The generic background/support information is made plant specific. In addition, each generic technical guideline is reviewed to determine the information needed to generate plant-specific procedures. An emergency operating procedures writer's guide is developed prior to generating these procedures from the Westinghouse Owners Group technical guidelines. In summary, the emergency operating procedures are developed using the Westinghouse Owners Group technical guidelines, plant-specific background/support information, and the emergency operating procedures writer's guide.

Each completed procedure is subjected to a two-part review in accordance with the emergency operating procedures verification program. The first part of this review verifies verbal accuracy, while the second part ensures technical accuracy. Upon completion of the verification program, the procedures are tested in accordance with the emergency operating procedures validation program. The validation program requires that all procedures be validated by at least one of the following (in order of preference for validation): using the plant-specific simulator; actual testing on plant systems; or detailed review.

Use of the Westinghouse Owners Group technical guidelines and supplemental information, in accordance with the emergency operating procedures action plan, meets the intent of NUREG-0899, items I.C.1 and I.C.8, as clarified by NUREG-0737, Supplement 1.

G. Temporary Procedures

Temporary procedures are issued, as required, to provide detailed instructions for specific jobs that are of a specific duration.

13.5.2.2 Maintenance Procedures

The VEGP maintenance procedures are divided into the following categories:

A. Generic Procedures

These procedures, which are general in nature, cover maintenance activities that can be performed on many different pieces of equipment or instrumentation; i.e., vibration analysis, inspections, insulation resistance checks, calibrations, functional tests, etc.

B. Specific Maintenance Activity Procedures

These procedures give instructions for service, repair, calibration, or functional testing of specific pieces of equipment, components, or instrumentation.

C. Surveillance Procedures

These procedures describe how to perform testing functions or parts replacement, as required, to verify the operational readiness of structures, systems, components, and instrumentation.

D. Special Procedures

Procedures for the maintenance, repair, calibration and functional testing of health physics, fire protection, and security equipment are also provided.

The maintenance department has responsibility for work performed in accordance with these procedures. Plant maintenance and modifications are also governed by administrative procedures.

13.5.2.3 Other Procedures and Program

13.5.2.3.1 Procedures

Other procedures are provided in the following areas:

A. Health Physics Procedures

Plant radiation protection procedures are designed to limit and control radiation exposure and the spread of contamination, as well as to meet the requirements of 10 CFR 20 and the as low as reasonably achievable philosophy.

The procedures describe rules, and practices or guidelines, for personnel protection, radiation surveys, decontamination, handling of radioactive or contaminated materials, and implementation of the as low as reasonably achievable program. The health physics department has responsibility for implementing these procedures and for ensuring the compliance of the plant staff with them.

B. Laboratory Procedures

These procedures describe rules, practices, or guidelines for tests, analyses, additions, or dilutions with respect to plant chemistry and radiochemistry. The health physics and chemistry laboratories are responsible for performing these activities.

C. Refueling Procedures

These procedures provide for preparation and performance of refueling operations. They include procedures to disassemble components, refueling equipment preuse maintenance and checkouts, and methods and limits for performing refueling operations.

D. Emergency Plan Implementation Procedures

These procedures provide rules and practices and designate responsibility and authority for classifying emergencies and responses to such emergencies. The plant staff has the responsibility to follow these procedures. Procedures in this section implement the emergency plan.

13.5.2.3.2 Program

The Process Control Program (PCP) is described below.

A. The PCP shall contain the current formulas, sampling, analyses, tests, and determinations to be made to ensure that processing and packaging of solid radioactive wastes based on demonstrated processing of actual or simulated wet solid wastes will be accomplished in such a way as to assure compliance with 10 CFR Parts 20, 61, and 71, State regulations, burial ground requirements, and other requirements governing the disposal of solid radioactive waste.

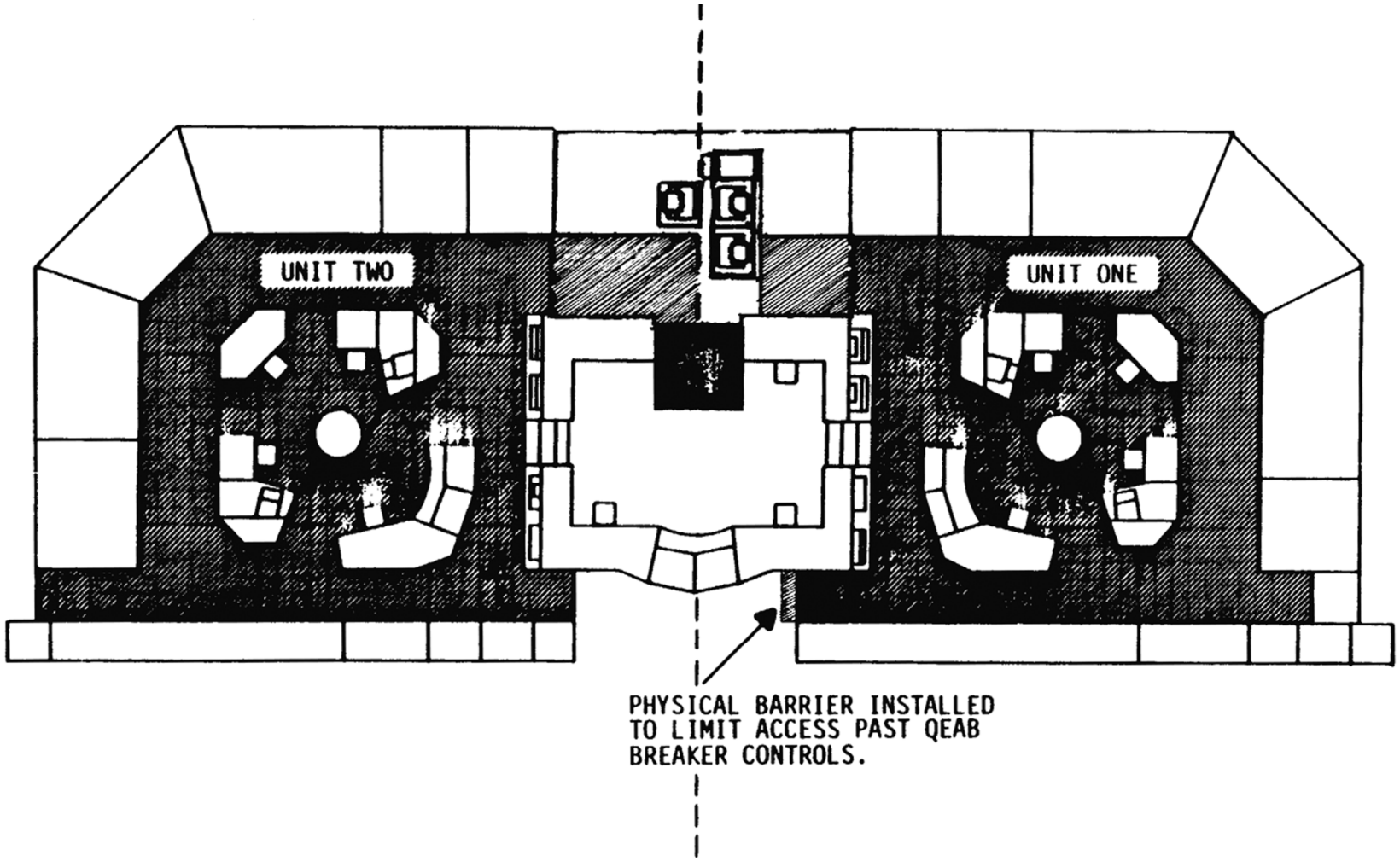
B. Licensee-initiated changes to the PCP:

1. Shall be documented and records of reviews performed shall be retained as required by paragraph 13.7.1. This documentation shall contain:

   a. Sufficient information to support the change together with the appropriate analyses or evaluations justifying the change(s), and

   b. A determination that the change will maintain the overall conformance of the solidified waste product to existing requirements of Federal, State, or other applicable regulations.

2. Shall become effective after review and acceptance by the PRB and the approval of the vice president-Vogtle.



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FIGURE 13.5.1–1

13.6 INDUSTRIAL SECURITY

A description of the physical security program for VEGP Units 1 and 2 has been provided to the Nuclear Regulatory Commission as a separate part of the application which is withheld from public disclosure pursuant to paragraph 10 CFR 2.790(d), Rules of Practice.

13.7 PLANT RECORDS

Records documenting the quality of the design, construction, testing, operation, maintenance, and modification of the Vogtle Electric Generating Plant (VEGP) are maintained at the plant site. The records are stored and maintained in accordance with the requirements of Criteria XVII of Appendix B of 10 CFR 50 as described in the SNC Quality Assurance Topical Report (QATR).

13.7.1 RECORD RETENTION

In addition to the applicable record retention requirements of Title 10, Code of Federal Regulations, the following records shall be retained for at least the minimum period indicated.

The following records shall be retained for at least 5 years:

A. Records and logs of plant operation covering time interval at each power level;

B. Records and logs of principal maintenance activities, inspections, repair, and replacement of principal items of equipment related to nuclear safety;

C. All reportable events;

D. Records of surveillance activities, inspections, and calibrations required by technical specifications;

E. Records of changes made to the procedures required by the QATR;

F. Records of radioactive shipments;

G. Records of sealed source and fission detector leak tests and results; and

H. Records of annual physical inventory of all sealed source material of record.

The following records shall be retained for the duration of the plant operating license:

A. Records and drawing changes reflecting plant design modifications made to systems and equipment described in the FSAR;

B. Records of new and irradiated fuel inventory, fuel transfers, and assembly burnup histories;

C. Records of radiation exposure for all individuals entering radiation control areas;

D. Records of gaseous and liquid radioactive material released to the environs;

E. Records of transient or operational cycles for those plant components identified in table 3.9.N.1-2;

F. Records of reactor tests and experiments;

G. Records of training and qualification for current members of the plant staff;

H. Records of inservice inspections performed pursuant to the VEGP inservice inspection program;

I. Records of quality assurance activities required by the FSAR;

J. Records of reviews performed for changes made to procedures or equipment or reviews of tests and experiments pursuant to 10 CFR 50.59;

K. Records of meetings of the PRB and SRB;

L. Records of the service lives of all hydraulic and mechanical snubbers required by the technical requirements manual, including the date at which the service life commences and associated installation and maintenance records;

M. Records of secondary water sampling and water quality;

N. Records of analyses required by the radiological environmental monitoring program that would permit evaluation of the accuracy of the analysis at a later date. This should include procedures effective at specified times and QA records showing that these procedures were followed; and

O. Records of reviews performed for change made to the offsite dose calculation manual (ODCM) and the process control program (PCP).

13.8 RISK INFORMED CATEGORIZATION AND TREATMENT

13.8.1 INTRODUCTION

On November 22, 2004, the NRC issued 10 CFR 50.69 that presented nuclear management with an opportunity to further enhance equipment reliability and plant safety by focusing on those critical structures, systems, and components (SSCs) with the highest safety significance. The rule broadly adjusts the scope of safety-related components that are subject to the existing NRC regulations. The implementation of this new rule is strictly voluntary on the part of each licensee.

Current NRC regulations define the plant equipment necessary to meet the deterministic regulatory basis as "safety-related." This equipment is subject to NRC special treatment regulations. These special treatment requirements go beyond normal commercial and industrial practices and include quality assurance (QA) requirements, qualification requirements, inspection and testing requirements, and Maintenance Rule requirements. Other plant equipment is classified as "nonsasfety-related'" and is typically not subject to special treatment requirements. For those components that are categorized as Low Safety Significant, 10 CFR 50.69 (b)(1) allows compliance with alternative requirements in lieu of the following special treatment requirements.

(i) 10 CFR part 21.

(ii) The portion of 10 CFR 50.46a(b) that imposes requirements to conform to Appendix B to 10 CFR part 50.

(iii) 10 CFR 50.49.

(iv) 10 CFR 50.55(e).

(v) The in-service testing requirements in 10 CFR 50.55a(f):

The in-service inspection and repair and replacement (with the exception of fracture toughness), requirements for ASME Class 2 and Class 3 SSCs in 10 CFR 50.55a(g); and the electrical component quality and qualification requirements in Section 4.3 and 4.4 of IEEE 279, and Sections 5.3 and 5.4 of IEEE 603-1991, as incorporated by reference in 10 CFR 50.55a(h).

(vi) 10 CFR 50.65, except for paragraph (a)(4).

(vii) 10 CFR 50.72.

(viii) 10 CFR 50.73.

(ix) Appendix B to 10 CFR part 50.

(x) The Type B and Type C leakage testing requirements in both Options A and B of Appendix J to 10 CFR part 50, for penetrations and valves meeting the following criteria:

(A) Containment penetrations that are either 1-in. nominal size or less, or continuously pressurized.

(B) Containment isolation valves that meet one or more of the following criteria:

(1) The valve is required to be open under accident conditions to prevent or mitigate core damage events;

(2) The valve is normally closed and in a physically closed, water-filled system;

(3) The valve is in a physically closed system whose piping pressure rating exceeds the containment design pressure rating and is not connected to the reactor coolant pressure boundary; or

(4) The valve is 1-in. nominal size or less.

(xi) Appendix A to part 100, Sections VI(a)(1) and VI(a)(2), to the extent that these regulations require qualification testing and specific engineering methods to demonstrate that SSCs are designed to withstand the safe shutdown earthquake and operating basis earthquake.

It should be noted that 10 CFR 50.69 does not replace the existing “safety-related” and “nonsafety-related” classification. Instead, 10 CFR 50.69 divides these classifications into two subcategories based on high or low safety significance, such that there are four categories of risk-informed safety class (RISC), as shown below:

RISC-1: safety-related SSCs that perform (high) safety significant functions.

RISC-2: nonsafety-related SSCs that perform (high) safety significant functions.

RISC-3: safety-related SSCs that perform low safety-significant functions.

RISC-4: nonsafety related SSCs that perform low safety-significant functions.

When applying alternative treatment, 10 CFR 59.69 requires that the licensee “shall ensure, with reasonable confidence, that RISC-3 SSCs remain capable of performing their safety-related functions under design basis conditions, including seismic conditions and environmental conditions and effects throughout their service life.”

Southern Nuclear received approval from the NRC to implement 10 CFR 50.69 at Vogtle 1 & 2 as outlined in reference 1.

13.8.2 SSC CATEGORIZATION

As outlined in the Safety Evaluation Report (SER), Plant Vogtle 1 & 2 will use the methodology outlined in NEI 00-04, 10 CFR 50.69 SSC Categorization Guidance, Revision 0, to categorize SSCs. For pressure retention components, consistent with the NRC SER, Plant Vogtle 1 & 2 will use the passive component categorization method as approved by the NRC for ANO-2 and as outlined in reference 2.

The SSC categorization process is outlined in plant procedures and associated instructions.

10 CFR 50.69 (f)(2) requires updating the UFSAR to reflect which systems have been categorized. The following table is revised as part of the periodic UFSAR update to reflect systems that have been categorized.

System Name	System Designator
Containment Spray	1206
Radiation Monitoring	1609

13.8.3 SSC TREATMENT

13.8.3.1 Treatment of Component Categories

The programs or processes that implement the special treatment requirements are revised to recognize that the special treatments no longer apply to RISC-3 components. The programs or processes either allow continued application of the special treatments or acceptable alternative treatments, as applicable, to provide reasonable confidence that these components would perform their safety-related function under design basis accidents.

The following information provides the general approach for applying treatment for the component categories:

A. RISC-1 Components

The purpose of treatment applied to RISC-1 SSCs is to maintain compliance with NRC regulations and to ensure that these SSCs can perform their risk-significant functions consistent with the categorization process. These components continue to receive existing treatment required by NRC regulations and associated Vogtle 1 & 2 implementing procedures.

Some safety-related components may be credited for performing functions that are beyond the design basis or for performing safety-related functions under conditions that are beyond the design basis. The Vogtle 1 & 2 PRA does not take credit for such functions unless there is a basis for confidence that the component will be able to perform the functions (e.g., demonstrated ability of the component to perform the functions under the specified conditions). If credit is taken for such function beyond that described above, the process described in paragraph 13.8.3.3 will be used to evaluate these risk-significant functions to the extent that they are not being treated under current programs.

B. RISC-2 Components

The purpose of treatment applied to RISC-2 SSCs is to maintain their ability to perform risk-significant functions consistent with the categorization process. These components will continue to receive any existing special treatment required by NRC regulations and Vogtle 1 & 2 programs. Additionally, the risk-significant functions of these components will receive consideration for enhanced treatment. This consideration is described in paragraph 13.8.3.2.

C. RISC-3 Components

These components may receive alternative treatments, in lieu of previous special treatments, as described in the affected programs.

D. RISC-4 Components

The treatment of these components is not subject to regulatory control.

E. Uncategorized Components

Until a component is categorized, it continues to receive the special treatment required by NRC regulations and associated Vogtle 1 & 2 implementing programs, as applicable.

13.8.3.2 Enhanced Treatment of RISC-2 SSCs

Nonsafety-related HSS components may perform risk-significant functions that are not addressed by the special treatment requirements in NRC regulations or current Vogtle 1 & 2 programs.

When a nonsafety-related component is categorized as HSS, Plant Vogtle determines whether enhanced treatment is warranted to enhance the reliability and availability of the component in support of its HSS function(s). In particular, Plant Vogtle evaluates the treatment applied to the component to ensure that the existing controls are sufficient to maintain the reliability and availability of the component in a manner that is consistent with its categorization. This process evaluates the reliability of the component, the adequacy of the existing controls, and the need for any changes. If changes are needed, additional controls are applied to the component. In addition, the component is placed under the Maintenance Rule monitoring program, if not already scoped in the program. Additionally, as provided in the Nuclear Quality Assurance Topical Report (QATR), nonsafety-related HSS components are subject to the enhanced QA program. These controls will be specifically targeted to ensure reliability of the critical attributes that resulted in the component being categorized as HSS. Components under these controls will remain nonsafety-related, but the enhanced treatments will be appropriately applied to give additional confidence that the component will be able to perform its HSS function(s) when demanded.

These identified processes provide reasonable confidence that HSS components will be able to perform their risk significant functions. The validation of functionality of HSS SSCs (safety-related SSCs for which existing special treatment does not provide the applicable level of confidence and nonsafety-related SSCs) will consist of a documented technical evaluation under the corrective action program to determine what enhanced treatment, if any, is warranted for these SSCs to provide reasonable confidence that the applicable risk significant functions will be satisfied. The performance of these SSCs will be monitored to provide reasonable confidence for their ongoing capability to perform their risk significant functions. The design control process will evaluate facility changes affecting the risk-significant functions of these SSCs.

13.8.4 REFERENCES

1. Safety Evaluation by the Office of Nuclear Reactor Regulation related to Amendment No. 173 to Renewed Facility Operating License NPF-68 and Amendment No. 155 to Renewed Facility Operating License NP-81, Docket Nos. 50-424 and 50-425, Vogtle Electric Generating Plants, Units 1 and 2, ML14237A034.
2. Safety Evaluation by the Office of Nuclear Reactor Regulation Request for Alternative AN02-R&R-004, Revision 1, Request to Use Risk-Informed Safety Classification and Treatment for Repair/Replacement Activities in Class 2 and 3 Moderate and High Energy Systems, Third and Fourth 10-Year In-service Inspection Intervals," Docket No. 50-368, Arkansas Nuclear One Plant Unit 2, ML090930246.